

Sai Bhandar

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🔗 github.io/sai-portfolio/

EDUCATION

Carnegie Mellon University

May 2028

B.S in Computer Science and Business Administration (GPA: 3.89 / 4.00)

Pittsburgh, PA

- Deans List (All Semesters) | Relevant Coursework: Data Structures and Algorithms, Theoretical Computer Science
- Extracurriculars: Product Management Club (President), Steel City Shershaah (Dancer and Finance Chair)

WORK EXPERIENCE

Instawork

June 2026 – August 2026

Product Operations Intern

San Francisco, CA

- Engineered automated data pipelines and Salesforce workflows for competitive intel leads engine, implementing deduplication, targeted enrichment, and quality-filtering logic, tripling contact rate and generating 5 demos
- Created system to gather competitive intelligence from worker-submitted data; validated through 10,000+ user campaign that generated 300+ quality submissions, 10 leads, and competitor insights for 20 account managers

Carnegie Mellon University - Wood Neuro Research Group

May 2026 – Present

Research Assistant

Pittsburgh, PA

- Built and tested Apple Watch and iOS research app with 5 participants, integrating HealthKit, secure authentication, and backend data pipelines to collect 20+ physiological biomarkers for a longitudinal pain prediction study
- Deployed Cloudflare-hosted backend server with Supabase, enabling remote participants to securely upload data

Carnegie Mellon University - 15-112 and 15-213

Aug 2025 – Present

Computer Science Teaching Assistant

Pittsburgh, PA

- 15-112: Led recitations and office hours for 400+ student introductory Python programming course, teaching recursion, iterative algorithms, data structures, debugging strategies, and code efficiency
- 15-213: Taught 300 student Computer Systems course, covering topics like C, assembly, VM, and concurrency

Capitol Interactive

June 2025 – Sep 2025

Software Engineer and Business Development Intern

Washington, DC

- Built 5 AR proof-of-concept experiences for Smithsonian's Apollo 11 exhibit, validating AR use for museum tours
- Implemented computer vision features through Unity, React Vision, and ARCore, including object detection and environment occlusion to enable realistic experiences, translating insights from 30+ interviews into product features

PROJECTS

Relief IQ | Python, Pandas, scikit-learn, Modal, React, TailwindCSS, OpenAI API

1st Place – CMU NOVA Hackathon (\$2,000); 1st Place – CMU × BNY Case Competition (\$4,200)

- Developed an AI-powered disaster response dashboard in 8 hours, integrating a scikit-learn Random Forest pipeline trained to create resource allocation plans based on data from 2 natural disasters
- Designed 5 AI and computer vision systems including aid demand modeling, AI support, and damage analysis

Sidequest | Unity 6, C#, Python, PyTorch, OpenCV, SAM 2, Grounding DINO, ONNX/Sentis

- Developed endless runner game driven by user's real vehicle motion, fusing GPS with 100 Hz IMU to detect motion, stops and starts across cars, buses, and trains; verified by 180+ tests
- Built two levels: prerecorded clips processed offline with SAM 2 and Grounding DINO, audited frame-by-frame, and live rear-camera inference with YOLOX and Fast-SCNN on-device via Unity Sentis for real-time object detection

Tric - AI Scheduling Agent | n8n, OpenAI API, React, JavaScript, Python, Supabase, HTML/CSS, Workflow Automation

- Spearheaded development of AI-powered scheduling MVP that automated availability collection, preference parsing, and dynamic rescheduling workflows based on 30+ medical student interviews
- Validated MVP through tests with 10 users achieving 50% retention, 60% task completion, and 15 minutes saved

Dynamic Memory Allocator | C, Systems Programming

- Doubled allocator utilization and throughput through footer elimination, mini-blocks, and free list management
- Constructed dynamic memory allocator in C using segregated free lists, block coalescing, and bounded best fit

TECHNICAL SKILLS

Languages: Python, C, C#, C++, JavaScript, SQL, HTML/CSS, R, JSON, SML, Assembly

Technologies: React, Git, Supabase, FastAPI, PyTorch, Django, n8n, OpenClaw, Clay, NodeJS, NextJS, Docker